

SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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ABSTRACT. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. AUXILIARY LEMMAS

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for falling factorials

Definition 1.2 (Falling factorials). *For integers x, n*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n-1)(n-2) \cdots (n-k+1) = \prod_{j=0}^{k-1} (n-j), & \text{if } k > 0, \\ 0, & \text{else.} \end{cases}$$

We utilize the following notation for central factorials

Definition 1.3 (Central factorials). *For integers n, k*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = n \left(n + \frac{k}{2} - 1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n + \frac{k}{2} - 1\right)_{k-1} = \prod_{j=0}^{k-2} \left(n + \frac{k}{2} - 1 - j\right)$ are falling factorials.

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [?].

Lemma 1.4 (Central Newton's formula). *For a function f defined on integers and arbitrary integers x, t*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t),$$

where $\delta^k f(t)$ denotes the k -th central finite difference

$$\delta^k f(t) = \sum_{j=0}^k (-1)^j \binom{k}{j} f\left(t + \frac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ are central factorials defined by (1.3).

Proof. See [?, p. 462], and [?, section 2, eq. (19)], and [?, section 6, eq. (21)]. □

We have

Lemma 1.5 (Newton's formula for powers). *For non-negative integers n, m , and an arbitrary integer t ,*

$$n^m = \sum_{k=0}^m \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proof. Special case of Lemma (1.4) for $f(n) = n^m$. □

We have

Lemma 1.6 (Central factorials binomial form). *For integers n and $k \neq 0$,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k(k-1)!} \binom{n + \frac{k}{2} - 1}{k-1}_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

because of the identity in falling factorials $\frac{(x)_n}{n!} = \binom{x}{n}$. □

We have

Lemma 1.7 (Central factorials binomial form decomposition). *For non-negative integers n, k ,*

$$\frac{n^{[k]}}{k!} = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} \right].$$

Proof. We have $\binom{a}{k} = \frac{a}{k} \binom{a-1}{k-1}$. Let $a = n + \frac{k}{2}$, then

$$\binom{n + \frac{k}{2}}{k} = \frac{n + \frac{k}{2}}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} + \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Thus,

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \binom{n + \frac{k}{2}}{k} - \frac{1}{2} \binom{n + \frac{k}{2} - 1}{k-1}.$$

By moving under common denominator, and applying binomial recurrence yields

$$\frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} = \frac{1}{2} \left[2 \binom{n + \frac{k}{2}}{k} - \binom{n + \frac{k}{2} - 1}{k-1} \right] = \frac{1}{2} \left[\binom{n + \frac{k}{2}}{k} + \binom{n + \frac{k}{2} - 1}{k} + \binom{n + \frac{k}{2} - 1}{k-1} - \binom{n + \frac{k}{2} - 1}{k-1} \right].$$

Thus, the statement follows. $\square$

We have

Lemma 1.8 (Generalized hockey-stick identity). *For arbitrary integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey-stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$

We have

Lemma 1.9 (Centered hockey-stick identity). *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j - t + \frac{k}{2} + r}{k} = \binom{n - t + \frac{k}{2} + r + 1}{k+1} - \binom{1 - t + \frac{k}{2} + r}{k+1}.$$

Proof. By setting $a = 1 - t + \frac{k}{2} + r$, and $b = n - t + \frac{k}{2} + r$ into Lemma (1.8) yields

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

We have

Lemma 1.10 (Multifold sums of zero powers). *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

- (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).
- (2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.
- (3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.
- (4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.
- (5) Similarly, for arbitrary integer r , by hockey-stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof. □

We follow the convention

Convention 1.11. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [?, p. 162], and in [?].

2. INTRODUCTION

In this manuscript, we derive formulas for sums of powers by combining central Newton's interpolation formula for powers (1.5) with hockey-stick identity (1.9). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [?, Alg. (11.2)]. More precisely, the main idea is to determine binomial basis of Newton's formula, in terms of variants of difference operators, for example

$$\left\{ \begin{array}{l} f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{d^k f(a)}{dx^k}. \end{array} \right.$$

We may observe that each variation of Newton's formula above has its binomial basis

$$\left\{ \begin{array}{l} \frac{(x)_n}{n!} = \frac{1}{n!} x(x-1)(x-2) \cdots (x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} = \frac{1}{n!} x(x+1)(x+2) \cdots (x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} = \frac{1}{n!} \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = \frac{x}{n} \binom{x+\frac{n}{2}-1}{n-1}. \end{array} \right.$$

This allows us to successfully combine Newton's formula with hockey-stick type identity, to find closed forms for sums of powers of integers. Thus, we focus on closed forms of powers sums, by utilizing central Newton's formula, and hockey-stick identity (1.9).

3. MAIN RESULTS

We start our discussion from Lemma (??), which yields backward Newton's interpolation formula for powers. For example,

$$\begin{aligned} n^3 &= 0 \binom{n-1}{0} + 1 \binom{n}{1} - 6 \binom{n+1}{2} + 6 \binom{n+2}{3}, \\ n^3 &= 1 \binom{n-2}{0} + 1 \binom{n-1}{1} + 0 \binom{n}{2} + 6 \binom{n+1}{3}, \\ n^3 &= 8 \binom{n-3}{0} + 7 \binom{n-2}{1} + 6 \binom{n-1}{2} + 6 \binom{n}{3}, \\ n^3 &= 27 \binom{n-4}{0} + 19 \binom{n-3}{1} + 12 \binom{n-2}{2} + 6 \binom{n-1}{3}. \end{aligned}$$

Thus, for powers we have identity, Thus,

Proposition 3.1 (Powers in central binomial form). *For integers n, t and $m \geq 0$*

$$\begin{aligned} n^m &= \frac{(n-t)^{[0]}}{0!} \delta^0 t^m + \sum_{k=1}^m \frac{n-t}{k} \binom{n+t+\frac{k}{2}-1}{k-1} \delta^k t^m \\ &= t^m + \sum_{k=1}^m (n-t) \binom{n-t+\frac{k}{2}-1}{k-1} \frac{\delta^k t^m}{k} \end{aligned}$$

Now we expand the brackets in central binomial form above,

$$n^m = t^m + \sum_{k=1}^m \frac{\delta^k t^m}{k} \left[n \binom{n-t+\frac{k}{2}-1}{k-1} - t \binom{n-t+\frac{k}{2}-1}{k-1} \right].$$

Hence, we get the formula for ordinary sums of powers.

Corollary 3.2 (Centered ordinary sums of powers). *For integers $t, m \geq 0, n \geq 0$,*

$$\Sigma^1 n^m = \sum_{j=1}^n t^m + \sum_{k=1}^m \frac{\delta^k t^m}{k} \left[\sum_{j=1}^n j \binom{j-t+\frac{k}{2}-1}{k-1} - t \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right]$$

Now we notice that,

Lemma 3.3 (Binomial decomposition). *For non-negative integers r, n, m*

$$n \binom{n+r}{m} = (m+1) \binom{n+r}{m+1} - (r-m) \binom{n+r}{m}.$$

Proof. By expanding the brackets yields,

$$\begin{aligned} n \binom{n+r}{m} &= (m+1) \binom{n+r}{m+1} - (r-m) \binom{n+r}{m} \\ &= m \binom{n+r}{m+1} + \binom{n+r}{m+1} - r \binom{n+r}{m} + m \binom{n+r}{m}. \end{aligned}$$

Recall the extraction property of binomial coefficients, that is,

$$\binom{n}{k+1} = \frac{n-k}{k+1} \binom{n}{k}.$$

Now we can notice that,

$$\binom{n+r}{m+1} = \frac{n+r-m}{m+1} \binom{n+r}{m},$$

by extraction. Thus,

$$n \binom{n+r}{m} = m \frac{n+r-m}{m+1} \binom{n+r}{m} + \frac{n+r-m}{m+1} \binom{n+r}{m} - r \binom{n+r}{m} + m \binom{n+r}{m}.$$

By moving binomial coefficient $\binom{n+r}{m}$ out of the brackets, we get,

$$\begin{aligned} n \binom{n+r}{m} &= \binom{n+r}{m} \left[m \frac{n+r-m}{m+1} + \frac{n+r-m}{m+1} - r + m \right] \\ &= \binom{n+r}{m} \left[(m+1) \frac{n+r-m}{m+1} - r + m \right] \\ &= n \binom{n+r}{m}. \end{aligned}$$

This completes the proof. □

Thus, by setting $n = j$, and $r = -t + \frac{k}{2} - 1$, and $m = k - 1$ into Proposition (??) yields central decomposition identity.

Corollary 3.4 (Central binomial decomposition). *For integers $j, t, k \geq 0$,*

$$j \binom{j - t + \frac{k}{2} - 1}{k - 1} = k \binom{j - t + \frac{k}{2} - 1}{k} + \left[t + \frac{k}{2} \right] \binom{j - t + \frac{k}{2} - 1}{k - 1}.$$

Proof. By binomial decomposition (??), we have,

$$\begin{aligned}
 j \binom{j-t+\frac{k}{2}-1}{k-1} &= (k-1+1) \binom{j-t+\frac{k}{2}-1}{k-1+1} - \left[-t + \frac{k}{2} - 1 - (k-1)\right] \binom{j-t+\frac{k}{2}-1}{k-1} \\
 &= k \binom{j-t+\frac{k}{2}-1}{k} - \left[-t - \frac{k}{2}\right] \binom{j-t+\frac{k}{2}-1}{k-1} \\
 &= k \binom{j-t+\frac{k}{2}-1}{k} + \left[t + \frac{k}{2}\right] \binom{j-t+\frac{k}{2}-1}{k-1}.
 \end{aligned}$$

This completes the proof. $\square$

Thus, we have the relation for centered sums of powers,

$$\begin{aligned}
 \Sigma^1 n^m &= \sum_{j=1}^n t^m + \sum_{k=1}^m \frac{\delta^k t^m}{k} \left[\sum_{j=1}^n \left\{ k \binom{j-t+\frac{k}{2}-1}{k} + \left[t + \frac{k}{2}\right] \binom{j-t+\frac{k}{2}-1}{k-1} \right\} \right. \\
 &\quad \left. - t \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right].
 \end{aligned}$$

By rearranging it, we get,

$$\begin{aligned}
 \Sigma^1 n^m &= \sum_{j=1}^n t^m + \sum_{k=1}^m \frac{\delta^k t^m}{k} \left[\left\{ k \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k} + \left[t + \frac{k}{2}\right] \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right\} \right. \\
 &\quad \left. - t \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right] \\
 &= \sum_{j=1}^n t^m + \sum_{k=1}^m \frac{\delta^k t^m}{k} \left[k \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k} + \frac{k}{2} \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right].
 \end{aligned}$$

Therefore, formula for centered sums of powers follows.

Proposition 3.5 (Centered decomposition of power sums). *For integers $m, n \geq 0$, and an arbitrary integer t ,*

$$\Sigma^1 n^m = \sum_{j=1}^n t^m + \sum_{k=1}^m \delta^k t^m \left[\sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k} + \frac{1}{2} \sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k-1} \right].$$

Let us recall generalized hockey stick identity. That is, Therefore, by setting $a = -t + \frac{k}{2}$ and $b = n - t - \frac{k}{2} - 1$ yields

Proposition 3.6 (Centered hockey stick identity). *For integers n, j, t, k ,*

$$\sum_{j=1}^n \binom{j-t+\frac{k}{2}-1}{k} = \sum_{a=-t+\frac{k}{2}}^{n-t-\frac{k}{2}-1} \binom{a}{k} = \binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1}$$

Thus, closed form of centered sums of powers yields

Theorem 3.7 (Closed form of centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and arbitrary integer t ,*

$$\begin{aligned} \Sigma^1 n^m &= \sum_{j=1}^n t^m \\ &+ \sum_{k=1}^m \delta^k t^m \left[\left(\binom{n-t+\frac{k}{2}}{k+1} - \binom{-t+\frac{k}{2}}{k+1} \right) + \frac{1}{2} \left(\binom{n-t+\frac{k}{2}}{k} - \binom{-t+\frac{k}{2}}{k} \right) \right]. \end{aligned}$$

Let $a = n - t + \frac{k}{2}$, then,

$$\begin{aligned} &\left(\binom{a}{k+1} - \binom{a-n}{k+1} \right) + \frac{1}{2} \left(\binom{a}{k} - \binom{a-n}{k} \right) \\ &= \left(\binom{a}{k+1} - \binom{a-n}{k+1} + \frac{1}{2} \binom{a}{k} - \frac{1}{2} \binom{a-n}{k} \right) \\ &= \frac{1}{2} \left(2 \binom{a}{k+1} - 2 \binom{a-n}{k+1} + \binom{a}{k} - \binom{a-n}{k} \right) \\ &= \frac{1}{2} \left(\binom{a}{k+1} + \binom{a}{k+1} - \binom{a-n}{k+1} - \binom{a-n}{k+1} + \binom{a}{k} - \binom{a-n}{k} \right). \end{aligned}$$

By binomial recurrence $\binom{a+1}{k+1} = \binom{a}{k} + \binom{a}{k+1}$

$$\begin{aligned} &\frac{1}{2} \left(\binom{a}{k+1} + \binom{a}{k+1} - \binom{a-n}{k+1} - \binom{a-n}{k+1} + \binom{a}{k} - \binom{a-n}{k} \right) \\ &= \frac{1}{2} \left(\left[\binom{a}{k+1} + \binom{a}{k} \right] + \binom{a}{k+1} - \binom{a-n}{k+1} - \left[\binom{a-n}{k+1} - \binom{a-n}{k} \right] \right) \\ &= \frac{1}{2} \left(\binom{a+1}{k+1} + \binom{a}{k+1} - \binom{a-n}{k+1} - \binom{a-n+1}{k+1} \right) \\ &= \frac{1}{2} \left(\left[\binom{a+1}{k+1} + \binom{a}{k+1} \right] - \left[\binom{a-n}{k+1} + \binom{a-n+1}{k+1} \right] \right) \end{aligned}$$

Therefore, by setting $a = n - t + \frac{k}{2}$, we get simplified formula for centered sum of powers.

Proposition 3.8 (Simplified centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and an arbitrary integer t*

$$\begin{aligned} \Sigma^1 n^m &= \sum_{j=1}^n t^m \\ &+ \sum_{k=1}^m \frac{\delta^k t^m}{2} \left[\left(\binom{n-t+\frac{k}{2}+1}{k+1} + \binom{n-t+\frac{k}{2}}{k+1} \right) - \left(\binom{-t+\frac{k}{2}}{k+1} + \binom{-t+\frac{k}{2}+1}{k+1} \right) \right] \end{aligned}$$

Continuing similarly, we can derive formulas for multifold sums of powers by using centered hockey stick identity (3.6) repeatedly. For instance, for double sums of powers, we have

$$\begin{aligned} \Sigma^2 n^m &= t^m \Sigma^2 n^0 \\ &+ \sum_{k=1}^m \frac{\delta^k t^m}{2} \left[\sum_{j=1}^n \left(\binom{j-t+\frac{k}{2}+1}{k+1} + \binom{j-t+\frac{k}{2}}{k+1} \right) \right. \\ &\quad \left. - \left(\binom{-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 + \binom{-t+\frac{k}{2}+1}{k+1} \Sigma^1 n^0 \right) \right] \end{aligned}$$

Thus, by generalized hockey stick identity (??)

$$\begin{aligned} \sum_{j=1}^n \binom{j-t+\frac{k}{2}+1}{k+1} &= \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{-t+\frac{k}{2}+2}{k+2} \\ \sum_{j=1}^n \binom{j-t+\frac{k}{2}}{k+1} &= \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{-t+\frac{k}{2}+1}{k+2} \end{aligned}$$

By substituting closed forms above, we get

$$\begin{aligned} \Sigma^2 n^m &= t^m \Sigma^2 n^0 \\ &+ \sum_{k=1}^m \frac{\delta^k t^m}{2} \left\{ \left[\binom{n-t+\frac{k}{2}+2}{k+2} - \binom{-t+\frac{k}{2}+2}{k+2} \right] \right. \\ &\quad + \left[\binom{n-t+\frac{k}{2}+1}{k+2} - \binom{-t+\frac{k}{2}+1}{k+2} \right] \\ &\quad \left. - \left[\binom{-t+\frac{k}{2}}{k+1} \Sigma^1 n^0 + \binom{-t+\frac{k}{2}+1}{k+1} \Sigma^1 n^0 \right] \right\} \end{aligned}$$

By combining the common terms yields

$$\begin{aligned} \Sigma^2 n^m = t^m \Sigma^2 n^0 + \sum_{k=1}^m \frac{\delta^k t^m}{2} & \left\{ \left[\binom{n-t+\frac{k}{2}+2}{k+2} + \binom{n-t+\frac{k}{2}+1}{k+2} \right] \right. \\ & - \left[\binom{-t+\frac{k}{2}+2}{k+2} \Sigma^0 n^0 + \binom{-t+\frac{k}{2}+1}{k+2} \Sigma^0 n^0 \right] \\ & \left. - \left[\binom{-t+\frac{k}{2}+1}{k+1} \Sigma^1 n^0 + \binom{-t+\frac{k}{2}+0}{k+1} \Sigma^1 n^0 \right] \right\} \end{aligned}$$

Thus, formula for double centered sums of powers follows

Proposition 3.9 (Double centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and an arbitrary integer t*

$$\begin{aligned} \Sigma^2 n^m = t^m \Sigma^2 n^0 + \sum_{k=1}^m \frac{\delta^k t^m}{2} & \left\{ \left[\binom{n-t+\frac{k}{2}+2}{k+2} + \binom{n-t+\frac{k}{2}+1}{k+2} \right] \right. \\ & - \left[\binom{-t+\frac{k}{2}+2}{k+2} + \binom{-t+\frac{k}{2}+1}{k+2} \right] \Sigma^0 n^0 \\ & \left. - \left[\binom{-t+\frac{k}{2}+1}{k+1} + \binom{-t+\frac{k}{2}+0}{k+1} \right] \Sigma^1 n^0 \right\}. \end{aligned}$$

Therefore, by continuing similarly, we can derive formula for r -fold sums of powers by using centered hockey stick identity (3.6) repeatedly. We have

Theorem 3.10 (Multifold centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and an arbitrary integer t*

$$\begin{aligned} \Sigma^r n^m = t^m \Sigma^r n^0 + \sum_{k=1}^m \frac{\delta^k t^m}{2} & \left\{ \left[\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right] \right. \\ & \left. - \sum_{s=0}^{r-1} \left[\binom{-t+\frac{k}{2}+r-s}{k+r-s} + \binom{-t+\frac{k}{2}+r-s-1}{k+r-s} \right] \Sigma^s n^0 \right\}. \end{aligned}$$

Now we notice that

Proposition 3.11 (Multifold sum of zero powers). *For integers $r \geq 0$ and $n \geq 1$*

$$\Sigma^r n^0 = \binom{r+n-1}{r}$$

Proof. (1) Let $r = 0$, then $\Sigma^0 n^0 = n^0 = \binom{n-1}{0} = 1$, by definition (??).

(2) Let $r = 1$, then $\Sigma^1 n^0 = \sum_{k=1}^n \binom{k-1}{0} = \sum_{k=1}^n 1 = \binom{n}{1}$.

(3) Let $r = 2$, then $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \sum_{k=1}^n k = \binom{n+1}{2}$.

(4) Let $r = 3$, then $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.

(5) By induction over r and hockey stick identity $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$, the claim follows

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

□

Hence, by (3.10) and (3.11), binomial form of multifold sums of powers follows

Proposition 3.12 (Binomial form of multifold centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and an arbitrary integer t*

$$\begin{aligned} \Sigma^r n^m &= \binom{r+n-1}{r} t^m + \sum_{k=1}^m \frac{\delta^k t^m}{2} \left\{ \left[\binom{n-t+\frac{k}{2}+r}{k+r} + \binom{n-t+\frac{k}{2}+r-1}{k+r} \right] \right. \\ &\quad \left. - \sum_{s=0}^{r-1} \left[\binom{-t+\frac{k}{2}+r-s}{k+r-s} + \binom{-t+\frac{k}{2}+r-s-1}{k+r-s} \right] \binom{s+n-1}{s} \right\}. \end{aligned}$$

We may observe another remarkable result, by setting $t \rightarrow -t$ into formula above

Proposition 3.13 (Negated binomial centered sums of powers). *For integers $n \geq 0$, $m \geq 0$, and an arbitrary integer t*

$$\begin{aligned} \Sigma^r n^m &= (-1)^m \binom{r+n-1}{r} t^m \\ &\quad + (-1)^m \sum_{k=1}^m \frac{(-1)^k \delta^k t^m}{2} \left\{ \left[\binom{n+t+\frac{k}{2}+r}{k+r} + \binom{n+t+\frac{k}{2}+r-1}{k+r} \right] \right. \\ &\quad \left. - \sum_{s=0}^{r-1} \left[\binom{t+\frac{k}{2}+r-s}{k+r-s} + \binom{t+\frac{k}{2}+r-s-1}{k+r-s} \right] \binom{s+n-1}{s} \right\}. \end{aligned}$$

Proof. We have $\delta^k(-t)^m = (-1)^{m+k} \delta^k t^m$, and $(-t)^m = (-1)^m t^m$. Hence claim follows from (3.12). □

4. CONCLUSIONS

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>ValidateBackwardNewtonsFormulaForPowers.txt</code>	Validates Lem. (??)

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